22 A source X emits a sound wave of constant frequency *f*.

The wave is subsequently received at a stationary detector Y.

The frequency of the wave that is detected by Y is less than *f*.

What could be the reason for this?

A Between X and Y, the wave undergoes diffraction.

B Between X and Y, the wave undergoes reflection.

C X is moving away from Y.

D X is moving towards Y.